# ***Making Gas Filling Station Tasks Easy with a CRM Program(Developer Project)***


## **ABSTRACT:-**

Fuel retailing today is more than selling fuel. It focuses on giving customers a smooth, data-driven experience. This project introduces a CRM-based solution for gas stations. It manages daily tasks and creates smart, customer-focused workflows. The system integrates customer data, vehicle details, fuel usage patterns, and supplier information.

The CRM uses Salesforce features such as custom objects, automated workflows, and role-based security. This reduces data duplication and helps managers make decisions. It provides real-time insights on fuel and supplier performance. Dashboards show customer activity clearly.

This project shows how a CRM evolves from record-keeping to a strategic system. It speeds up service, reduces manual work, and builds customer trust. Overall, the CRM makes gas station management smarter, scalable, and easier for staff and customers.


## **OBJECTIVE:-**

The Gas Filling Station CRM project is designed to make fuel station management smoother and more customer-friendly. The main purpose of this system is to reduce difficulties in daily operations, ensure the safety of data, and create a better experience for everyone who uses it. All important information, such as customer details, vehicle numbers, supplier records, fuel stock, and station activities, is brought together into a single platform. This makes it easy for staff to access accurate data whenever needed, which also helps in serving customers faster and more efficiently. By keeping proper records of customers, their vehicles, and receipts, the system improves customer trust and satisfaction.

To make work more organized, role-based access is used. This means managers, executives, and salespersons each get secure access based on their responsibilities, which avoids confusion and ensures accountability. Another big advantage is automation. Tasks that are usually done manually—like updating buyer details, recording fuel entries, or handling payments—are managed by the CRM automatically.

The system also takes data protection seriously. With strong security rules and permissions, sensitive business and customer information stays safe from unauthorized access. Lastly, the project is built to grow with future needs. Features like detailed reports and interactive dashboards can be added easily, which will help in making better business decisions and improving station performance over time.

## Key Salesforce Technologies and Features:-

The Gas Station CRM project uses Salesforce, a cloud-based platform. It manages operations, customer data, and sales processes in one place. To build the system, the growth team utilized different Salesforce features:

- **Custom Objects:** These are user-defined tables for storing business-specific data.
- **Tabs:** Tabs provide quick navigation to objects. They let users work with data directly in the app."
- **Lightning App:** The development team customized the CRM application. It combines all objects, tabs, and utilities in one workspace.
- **Page Layouts:** Used to arrange fields and sections on records, ensuring data entry is simple.
- **Profiles:** Control what each user can see or do within Salesforce by defining access. The system integrates objects, fields, and features to support smooth operations.
- **Roles:** Establish the reporting structure and manage record visibility.
- **Permission Sets:** The system grants extra access beyond base profiles. Users gain flexibility without changing original settings.
- **Organization-Wide Access Settings:** Set the baseline access for records. They protect data while allowing collaboration
- **Reports and Dashboards**: Provide visual insights. They help users track fuel, sales, and customers.
- **Flows and Triggers:** Automate business processes. They enforce rules, preventing the deletion of important records.

This combination of Salesforce tools keeps the system secure, scalable, and user-friendly. It makes daily gas station operations more efficient.

## Project Implementation Steps:-

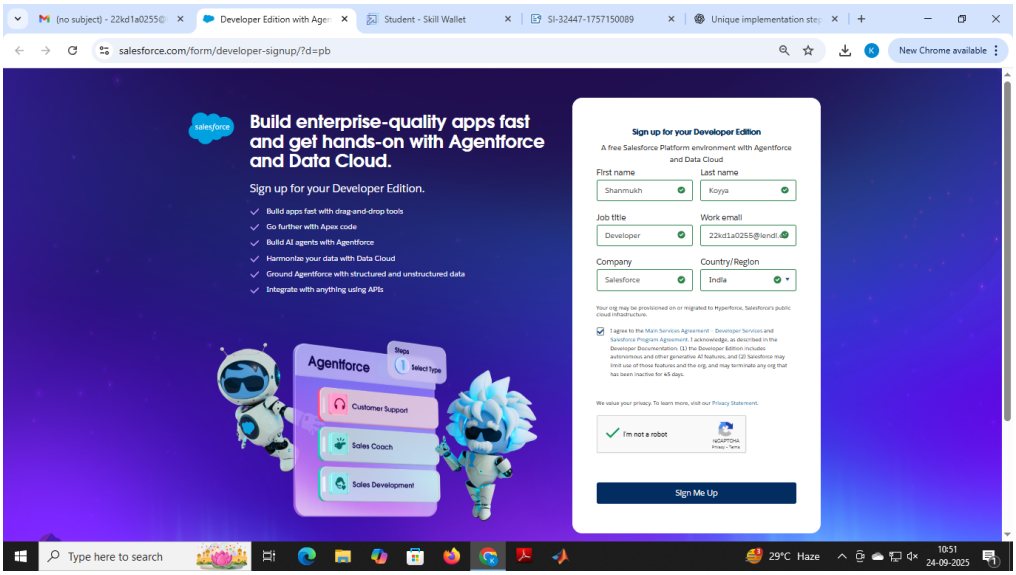
### 1. Setting Up a Salesforce Developer Org:-

**Purpose**:

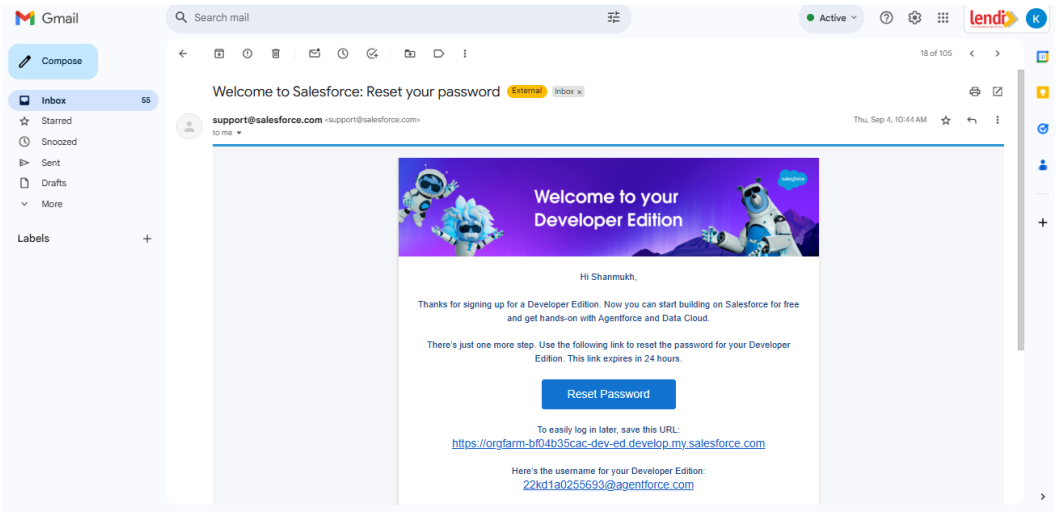
A Salesforce Developer Org is a secure environment to safely build, customize, and test apps, objects, automation, and user roles.

**Procedure:**

- Open developer.salesforce.com/signup.
- Fill in your details and create your free Developer account.




- Check your email, click the activation link, and set a password.



## 2. Creating Custom Objects:-

**Purpose:**

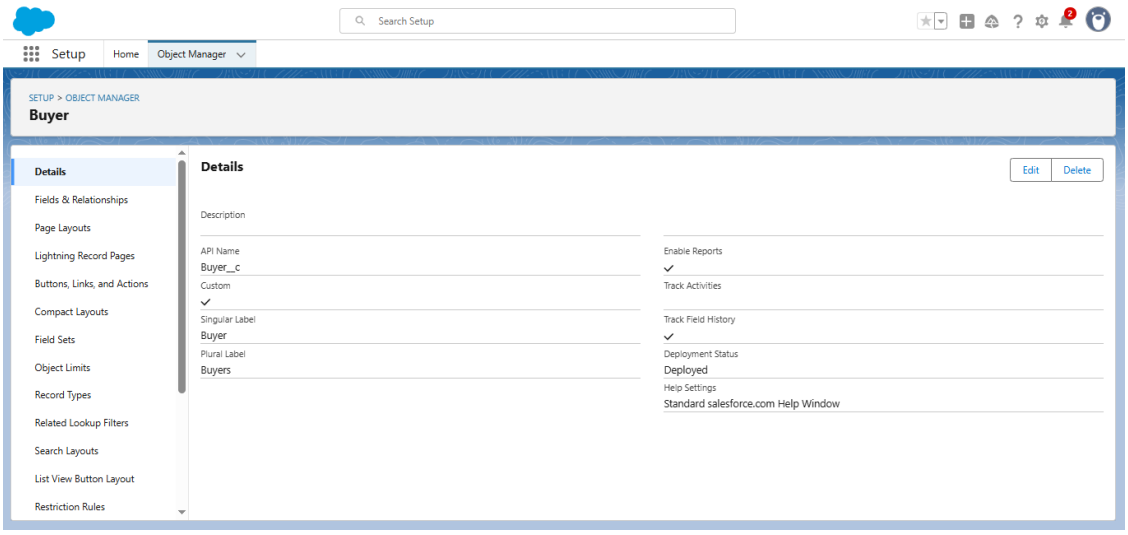
The development team uses fully customizable Salesforce custom objects to store business-specific data, designing fields, relationships, and layouts.

**Procedure:**

1. Navigate to Setup → Object Manager → New Object.
2. Enter the object label and API name.
3. Configure settings such as record name, data type, reporting, and field history tracking if required.
4. Add necessary fields and relationships to capture project-specific data.
5. Save the object to make it available in the organization.

## Objects Created in this Project:-

- **Buyer__c** – Stores customer information (name, contact, vehicle) and purchase history.
- **Fuel_Details__c** – Records each fuel transaction and links it with the buyer, supplier, and station.
- **Gas_Station__c** – Maintains station details, including stock levels, fuel usage, and availability.
- **Supplier__c** – Tracks suppliers and the total fuel supplied to gas stations.



## 3. Creating Custom Tabs:-

**Purpose:**

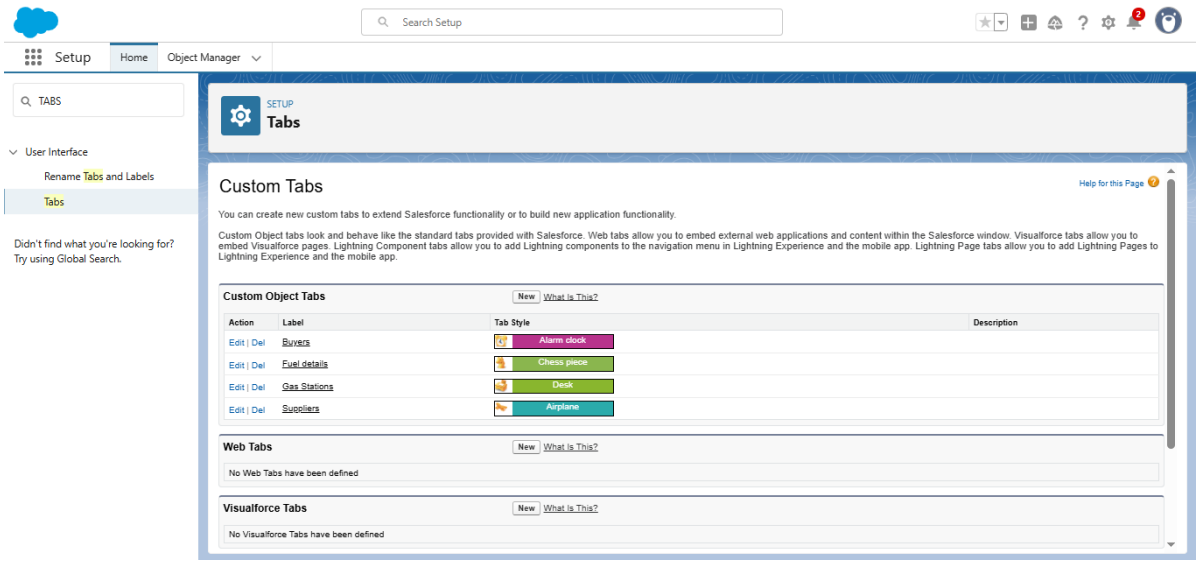
Salesforce tabs provide quick access to objects, simplifying navigation and allowing users to manage custom object records directly.

**Procedure:**

1. Go to Setup → Tabs → New Custom Object Tab.
2. Select the custom object (e.g., Buyer, Fuel Details, Gas Station, or Supplier).
3. Choose a tab style (icon and color) for easy identification.
4. Assign the tab to the correct profiles so only authorized users can view it.
5. Add the tab to the Lightning App for smooth navigation.

**Tabs Created in this Project:-**

- **Buyer Tab** is For adding and managing customer records.
- **Fuel Details Tab** is For recording and checking fuel transactions.
- **Gas Station Tab** is for monitoring stock levels, usage, and availability.
- **Supplier Tab** is for tracking suppliers and their deliveries.



## 4. Creating Custom Tabs:-

**Purpose:**

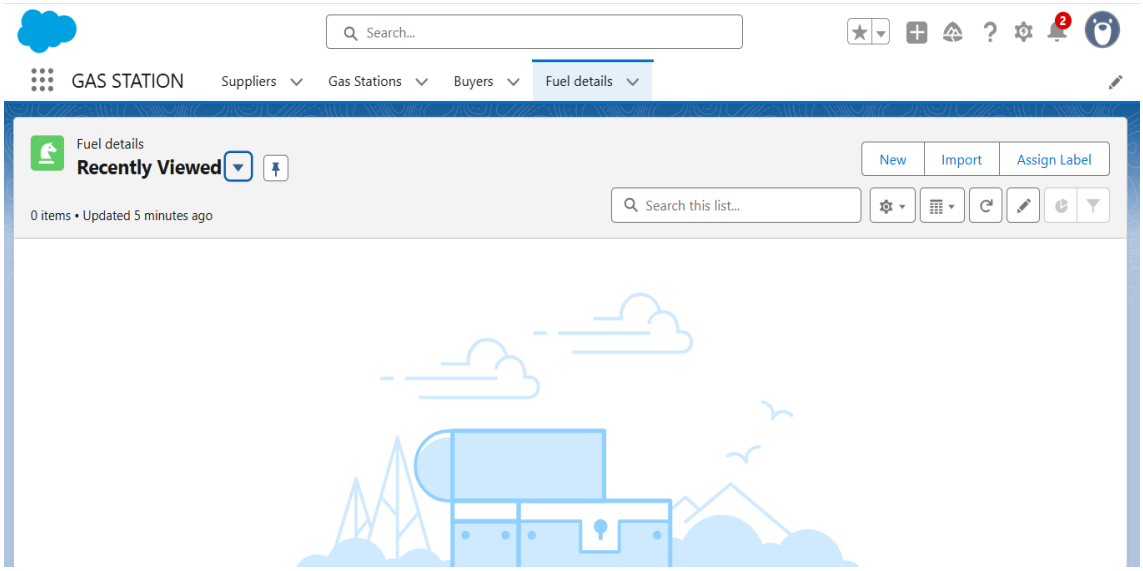
Salesforce tabs provide quick access to objects, making it easy to navigate and manage custom object records without searching menus.

**Procedure:**

1. Go to Setup → Tabs → New Custom Object Tab.
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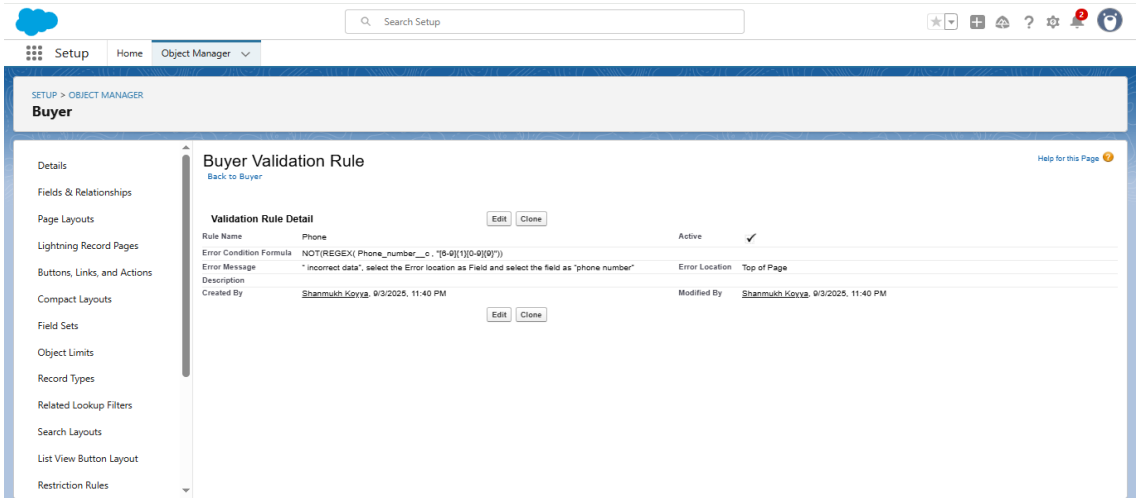


## 5. Fields to be created:-

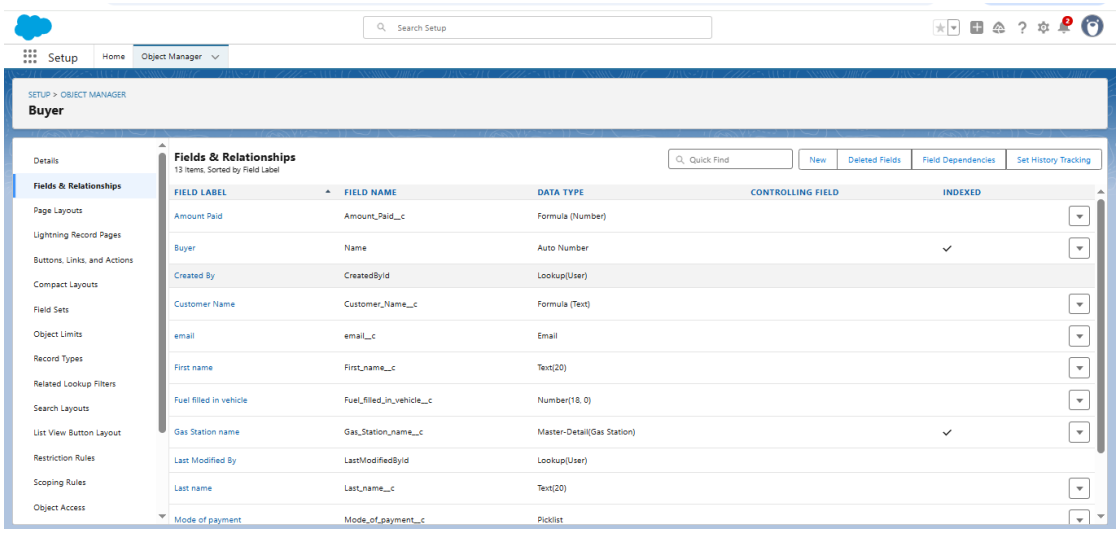
**Purpose:**

Custom fields let Salesforce objects capture specific business details, enabling the Gas Station CRM to track customers, transactions, stations, and suppliers accurately.

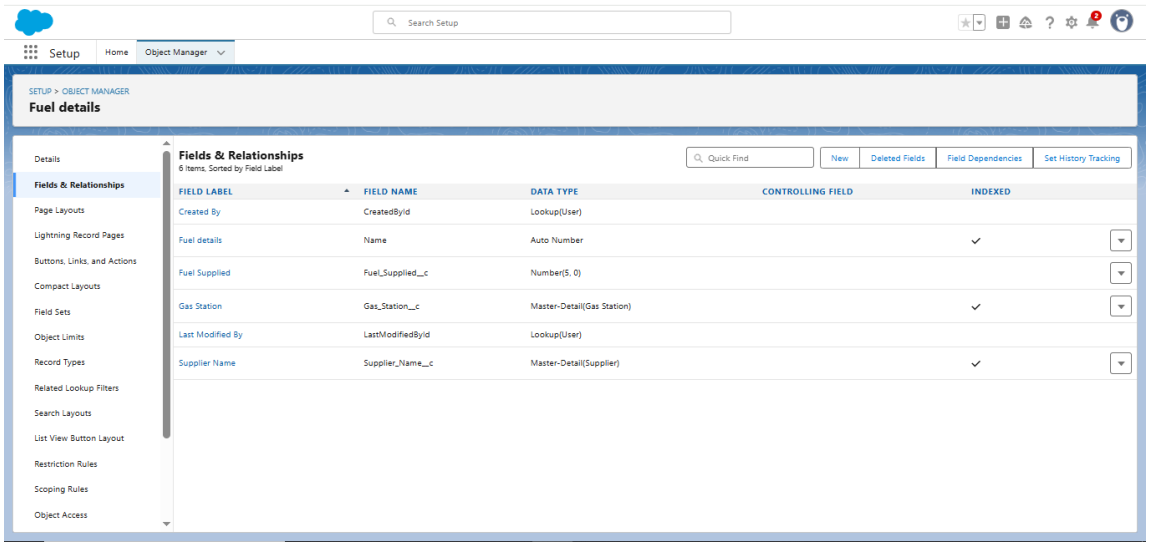
**Validation in buyer object** (phone number)



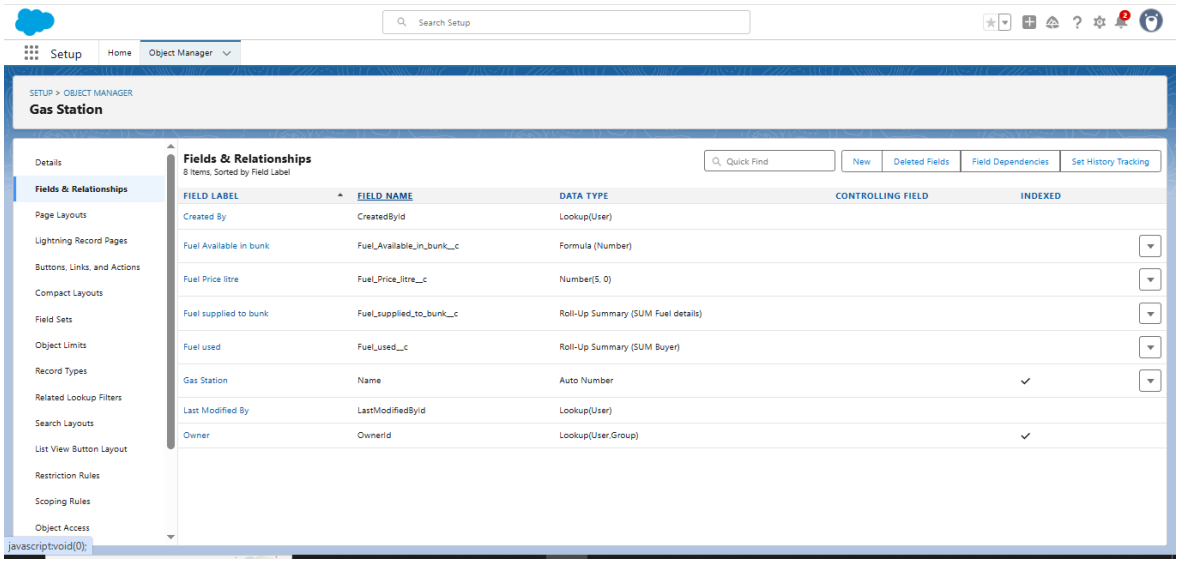

## Fields & Relationships in Buyer Object



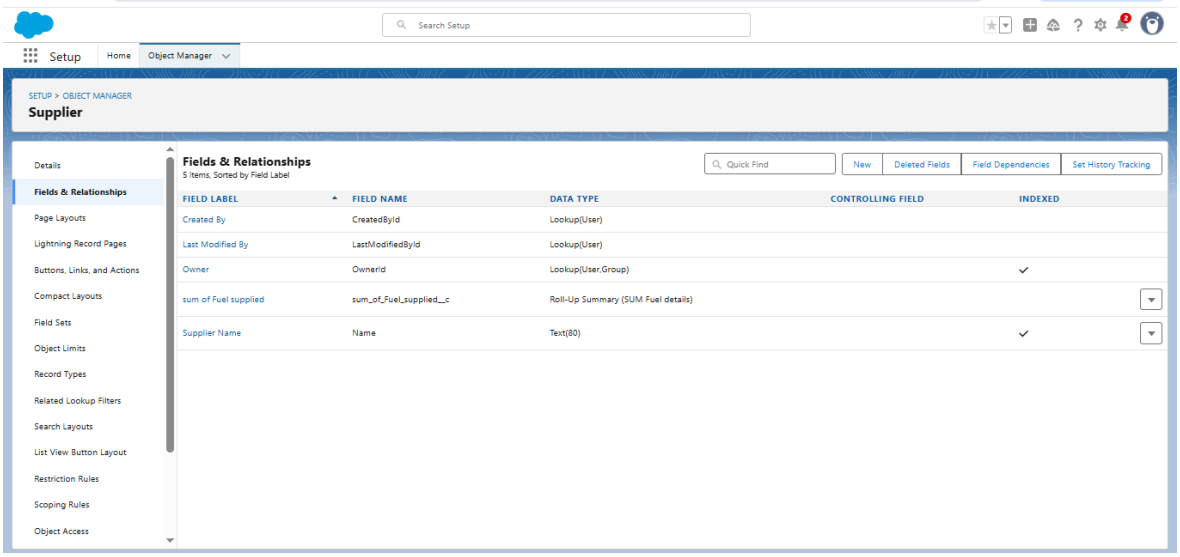
- **Fuel_details__c:** Fuel Supplied (Number), Supplier Name (Master-Detail), Gas Station (Master-Detail)




- **Gas_Station__c:** Gas_Station__c tracks fuel price, fuel supplied, fuel used, and fuel available.



- **Supplier__c:** Sum of Fuel Supplied (Roll-up Summary)



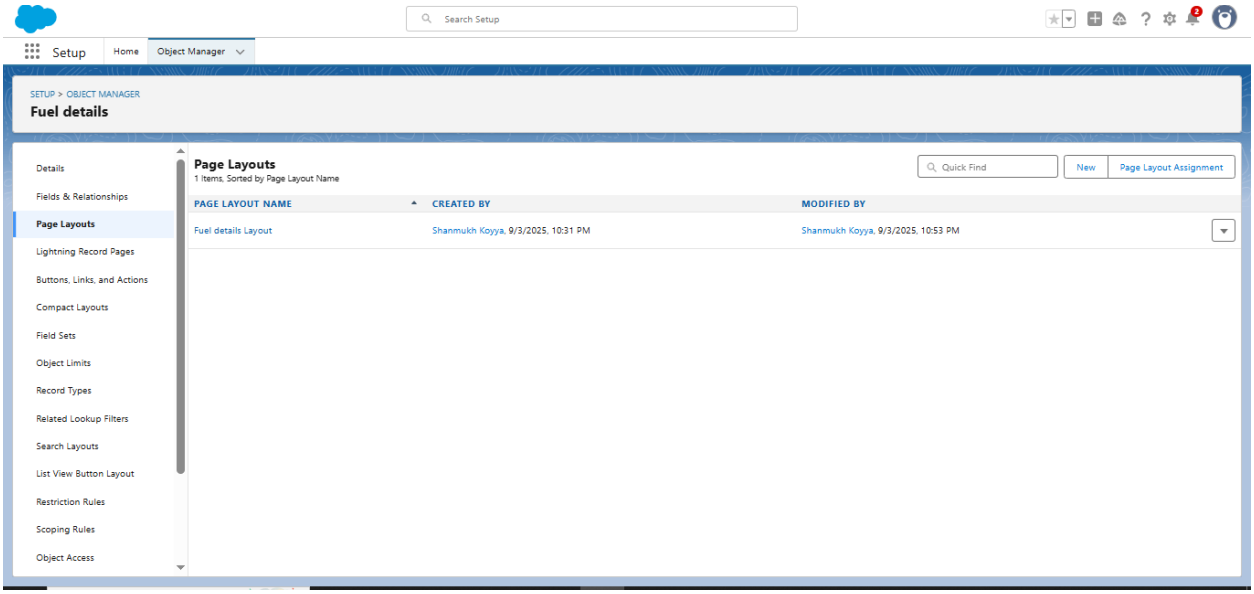

## 6. <u>Create Page Layout for Objects</u>:-

**Purpose:**

The administrator sets page layouts to organize fields, sections, and related lists, improving data entry speed, reducing errors, and aiding user navigation.

**Procedure:**

1. In Setup, go to Object Manager → [Object] → Page Layouts → New, then drag fields into sections like Contact, Fuel, Payments.
2. Save the layout and set it as the default view for the object.

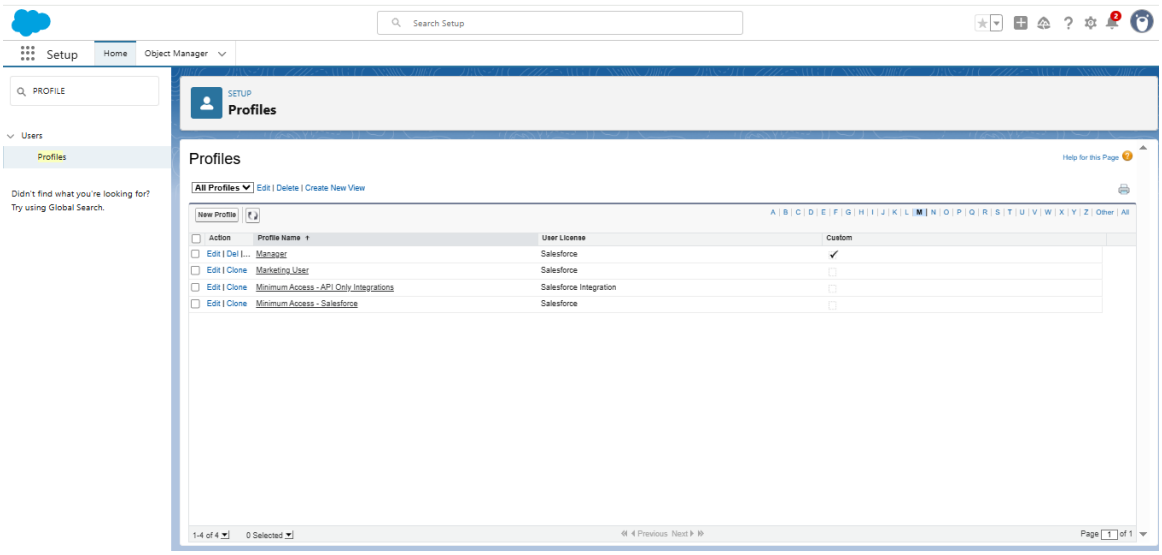



## 7. Create Profiles:-

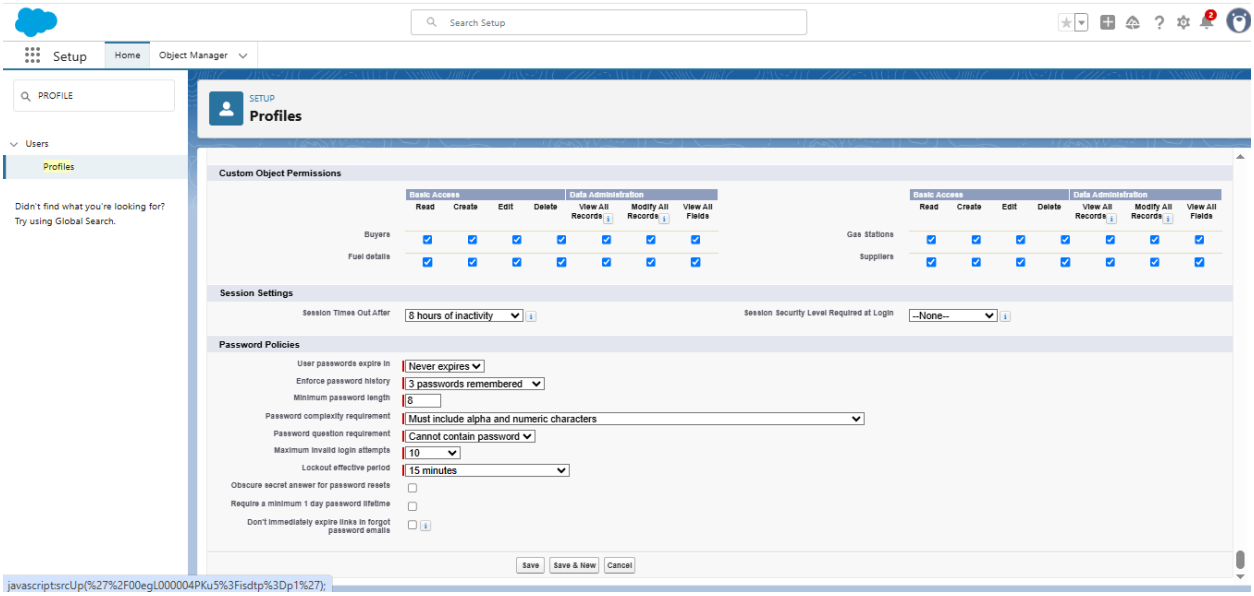
Profiles manage user permissions. Profiles determine which data and actions each role can access. They help maintain security and proper permissions.

### Steps Taken

- **The administrator** duplicated the Standard User profile to create the Manager profile. They used the Salesforce Platform User profile for Sales Executive and Salesperson roles.
- **Name & Save** – Assigned clear names and brief descriptions for each profile.
- **Set Permissions** – I set permissions and adjusted rights so roles handled only necessary tasks.
- **Enable App Access** – Granted visibility to the Gas Station Lightning App where required.




- The administrator edited each profile to set custom app and object permissions.
- Password Policies – Set rules… → fragment.
- After creating the profiles, → dangling phrase.

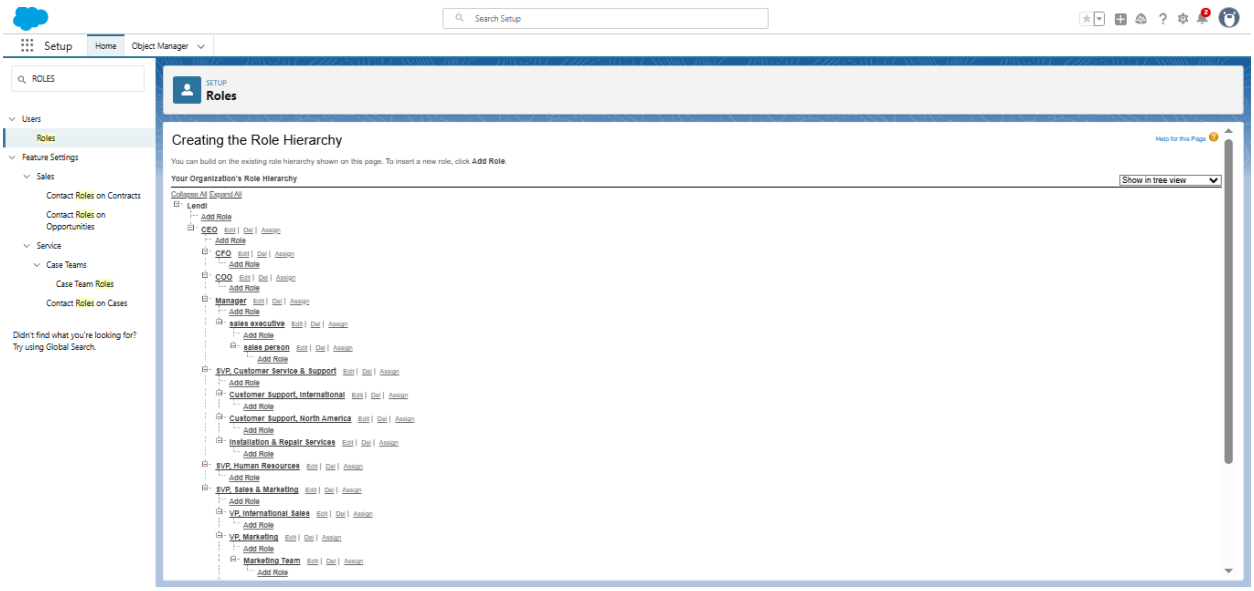


## 8. Creating Roles & Hierarchy:-

Roles decide **record-level access** and set up the reporting structure inside Salesforce.

**Steps Followed**

1. In Setup, go to Roles → Set Up Roles → Add Role. Enter “Manager” as the Role Label and save.
2. Add “Sales Executive” and “Salesperson” beneath Manager, assign profiles, and confirm hierarchy: Manager → Sales Executive → Salesperson.



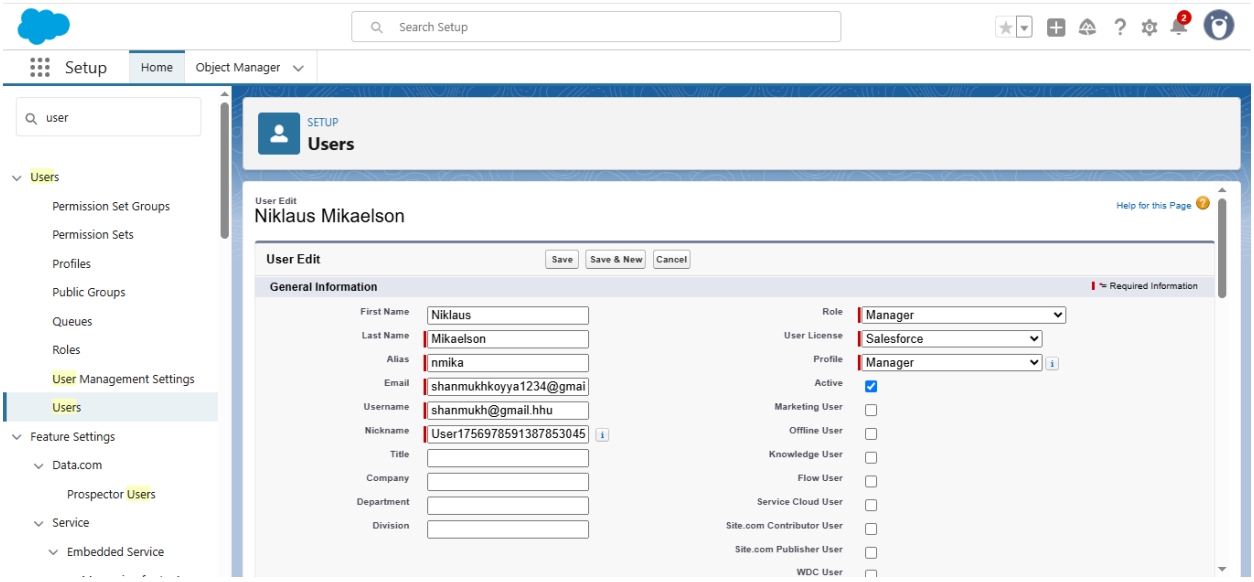

## 9. Creating users:-

**Purpose:-**

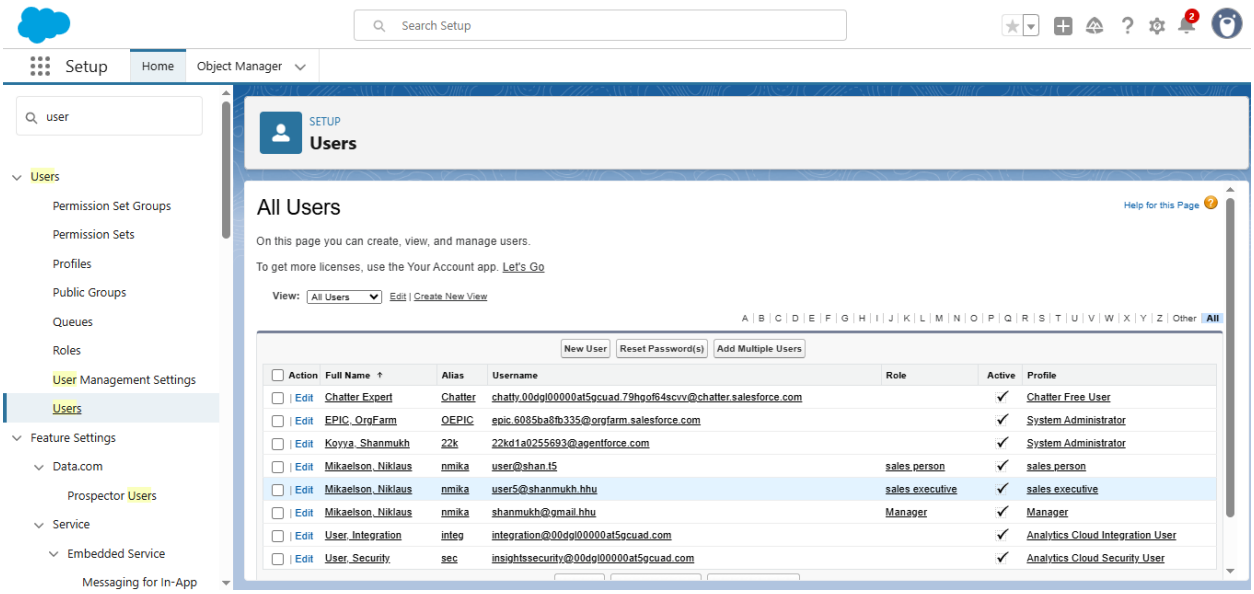
In Salesforce, each user has an individual account. They log in to perform tasks.The administrator assigns a role and profile to each user. This limits access to only necessary data and features. This approach enhances security and maintains a clear division of work.

**Procedure:-**

1. Go to Setup → Users → New User.
2. Enter user details such as Name, Email, Username, and Role.
3. Assign the appropriate Profile (e.g., Manager, Sales Executive, Sales Staff).
4. Save the user to activate their account.




The users appear in the application. Users hold roles such as Manager, Sales Executive, or Salesperson.



## 10. Create Permission Sets:-

**Purpose:-**

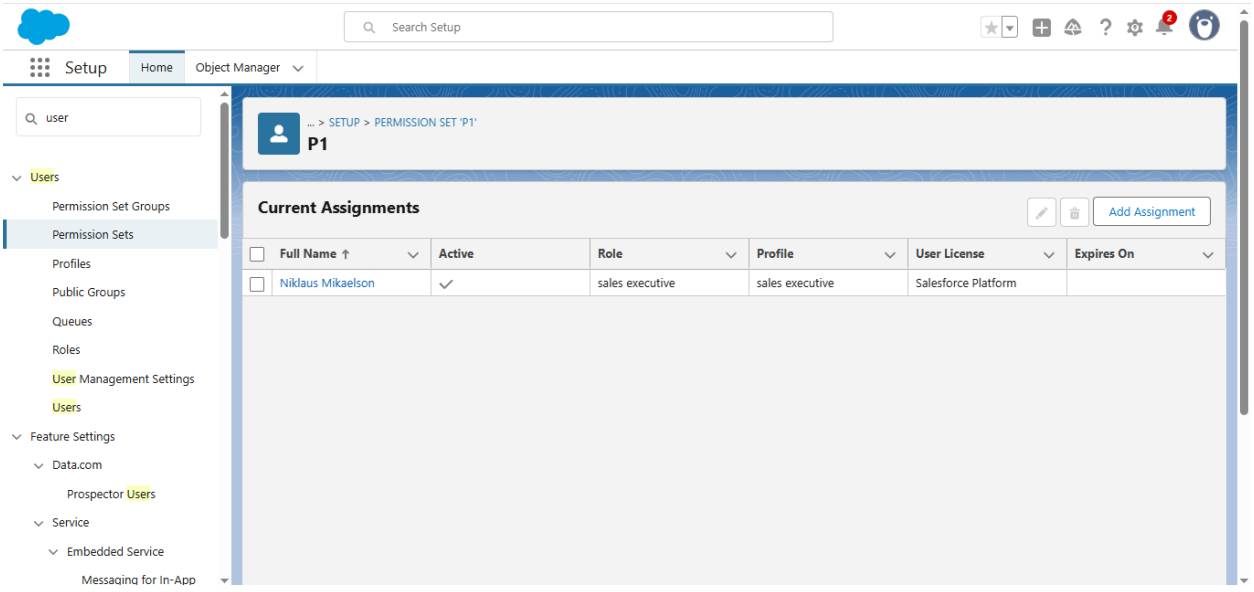
The administrator uses Permission Sets in Salesforce. They give users extra access without changing profiles. This provides flexibility when assigning permissions. Specific users can perform extra actions as needed.

**Procedure:-**

1. Go to Setup → Permission Sets → New.
2. Enter a Label and API Name for the permission set (e.g., P1).
3. Define the necessary object and field permissions to extend user access.
4. Save the permission set.
5. Assign it to the intended user(s) by selecting Manage Assignments → Add Assignments.

**Project Implementation:-**

- The administrator created the P1 Permission Set to provide extra access to certain objects.
- The administrator assigned the permission set to the Sales Executive. This gave them extra rights beyond their profile.




## 11. OWD (Organization-Wide Defaults):-

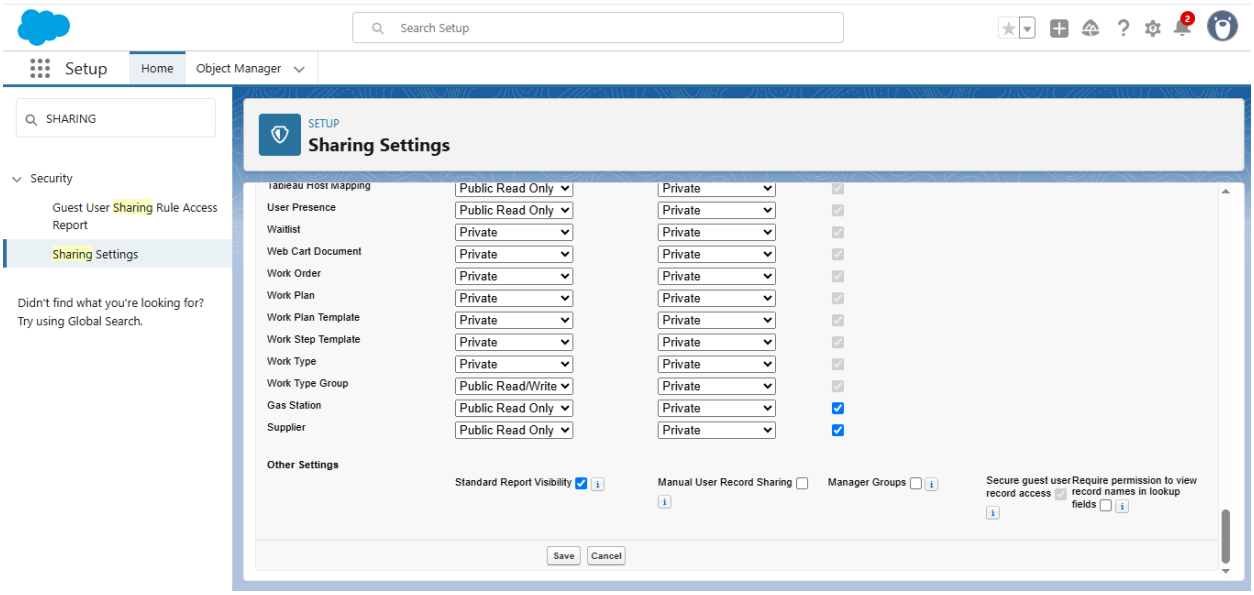
**Purpose:-**

Organization-Wide Defaults (OWD) set baseline access. They control what users can see in Salesforce. OWD ensures data security by limiting what users can view or edit. Collaboration is still allowed where appropriate.

**Procedure:-**

1. Go to Setup → Security → Sharing Settings.
2. Options include *Private, Public Read-Only, or Public Read/Write*.
3. Save the settings to apply them across the system.

This configuration balances security and collaboration by protecting sensitive data while allowing visibility where appropriate.



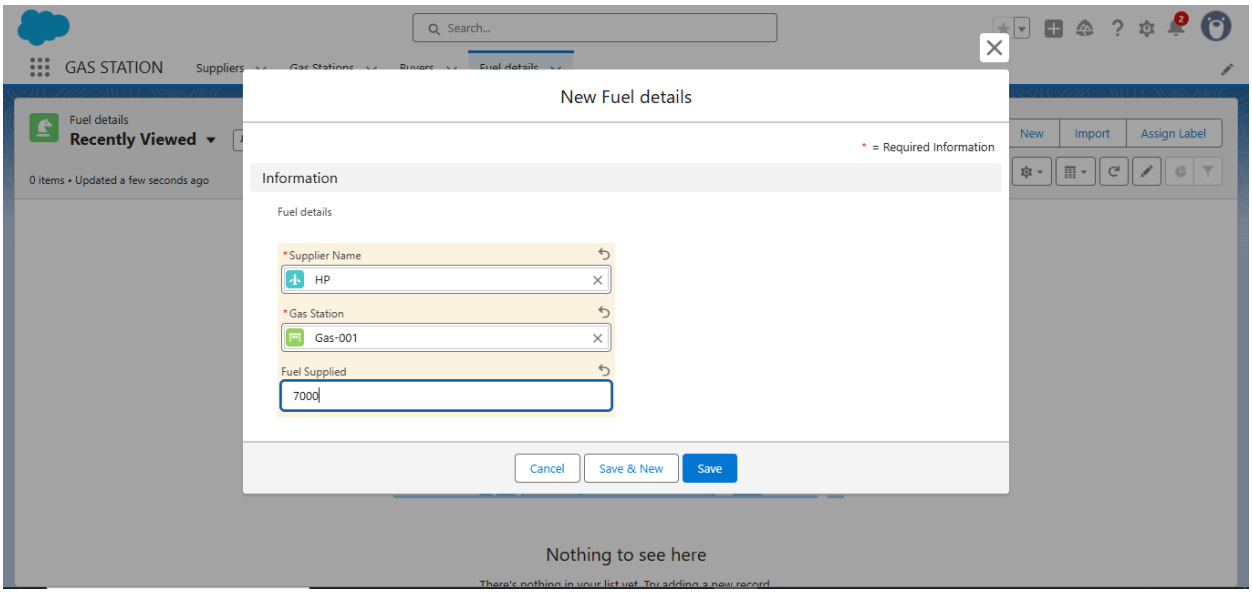

**12. Create, View, and Delete Records:-**

**Purpose:-**

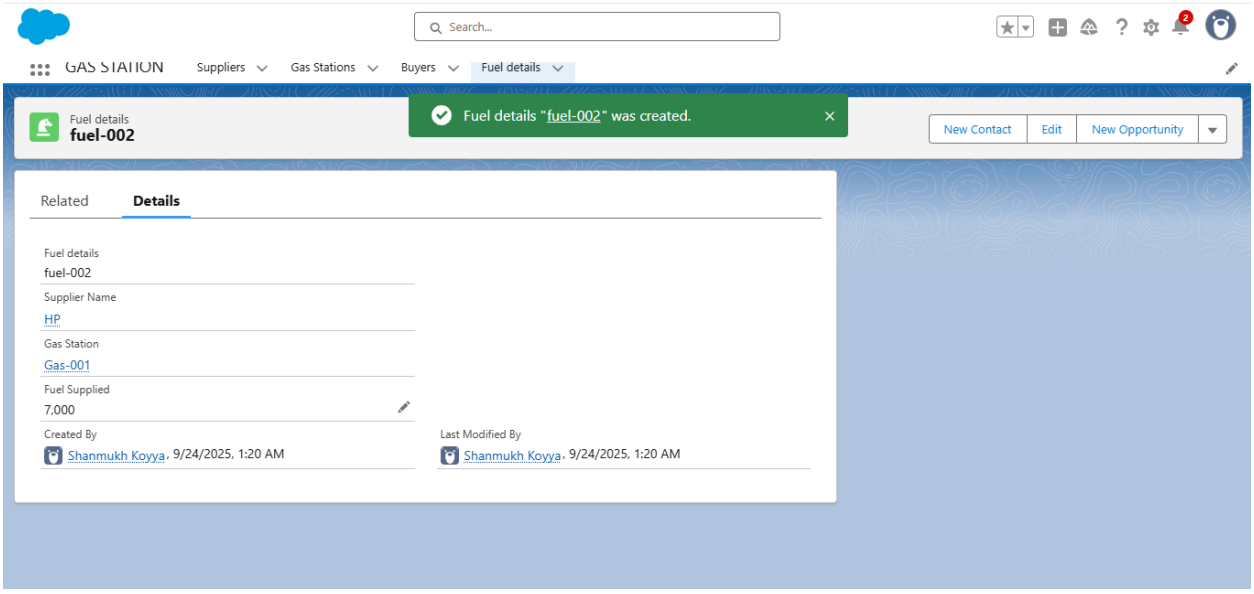
The ability to create, view, and delete records is central to user adoption in Salesforce. It shows CRM usage by team members. They complete daily tasks efficiently.

**Implementation in Project:-**

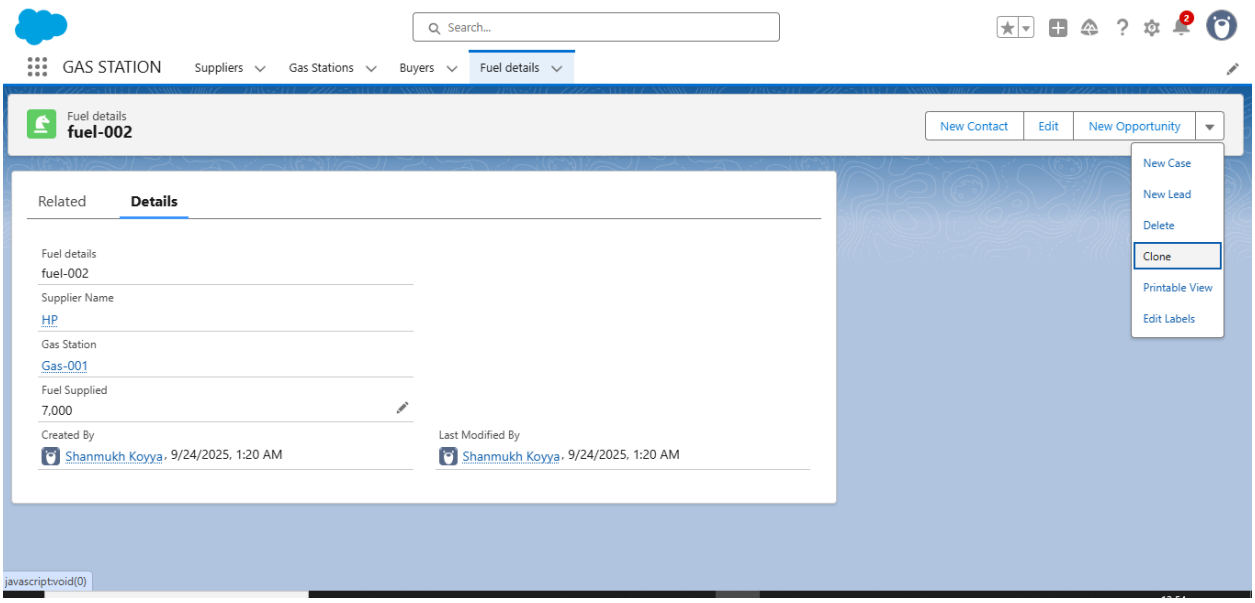
Roles, profiles, and permission sets were defined to provide appropriate access. Intuitive page layouts enabled easy record management, creation, viewing, and deletion, improving adoption among managers, sales executives, and staff.




## View Record




## Delete Record



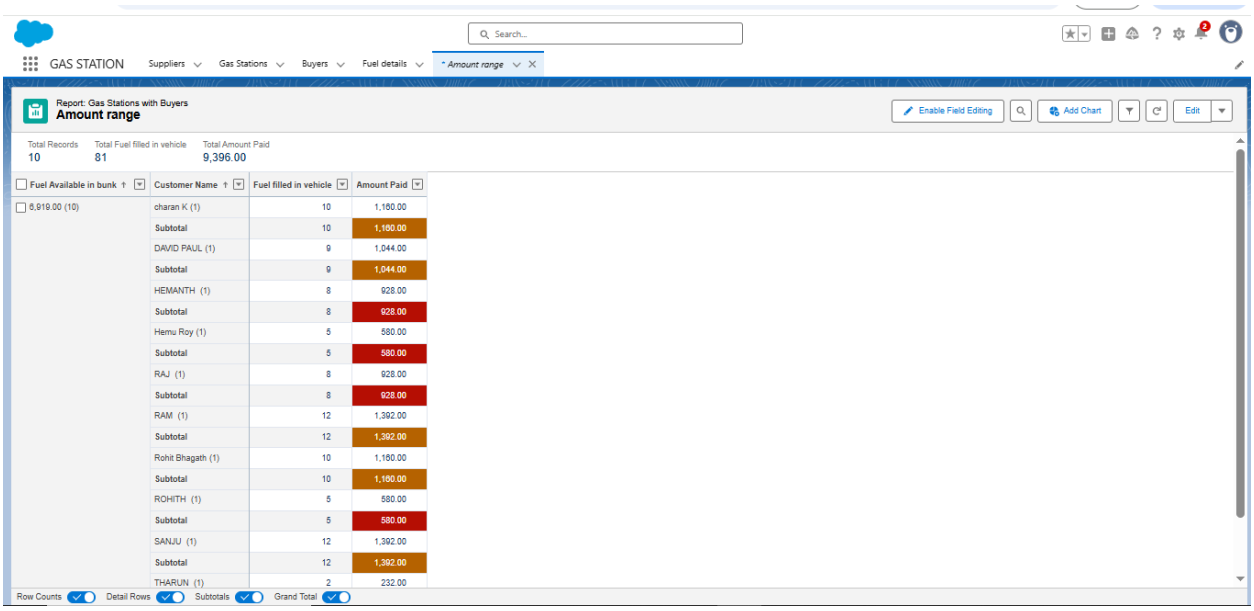

## 13. Creating Reports:-

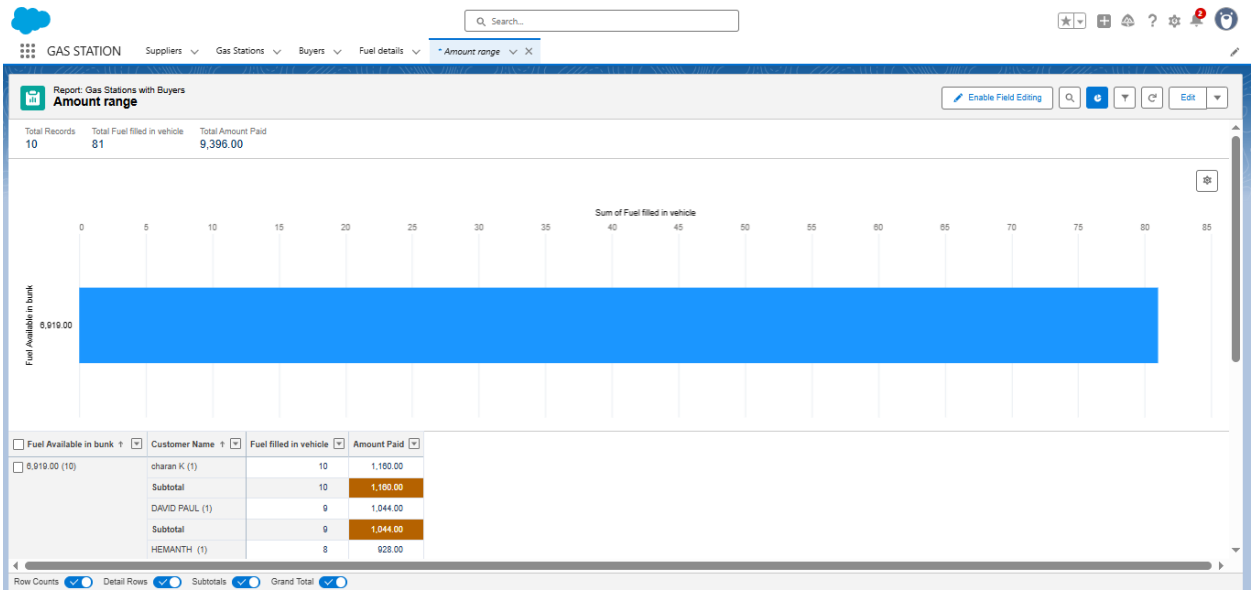
**Purpose:-**

Reports in Salesforce help analyze and present data in an organized format. They allow users to track performance, identify patterns, and make informed business decisions.

**Implementation in Project:-**

The administrator created custom reports to track fuel usage, customer activity, and sales. Automated deliveries kept stakeholders informed, supporting better performance monitoring and decision-making.




## 14. Creating Dashboards:-

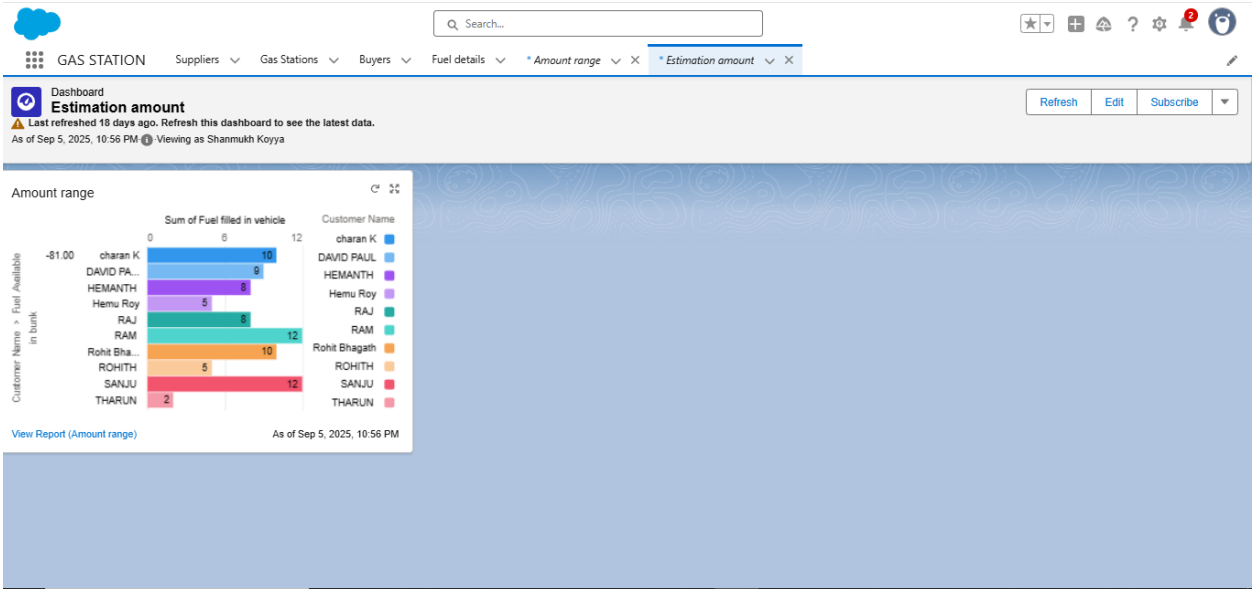
**Purpose:-**

Dashboards turn Salesforce reports into visual insights by using charts, graphs, and key metrics. They allow users to analyze performances and make decisions based on real-time

data.

**Implementation in Project:-**

- The manager designed dashboards to highlight fuel estimation, sales trends, and customer action.
- This gave managers and sales executives a clear view of operations.
- Dashboards simplified complex data through visuals and improved business monitoring.



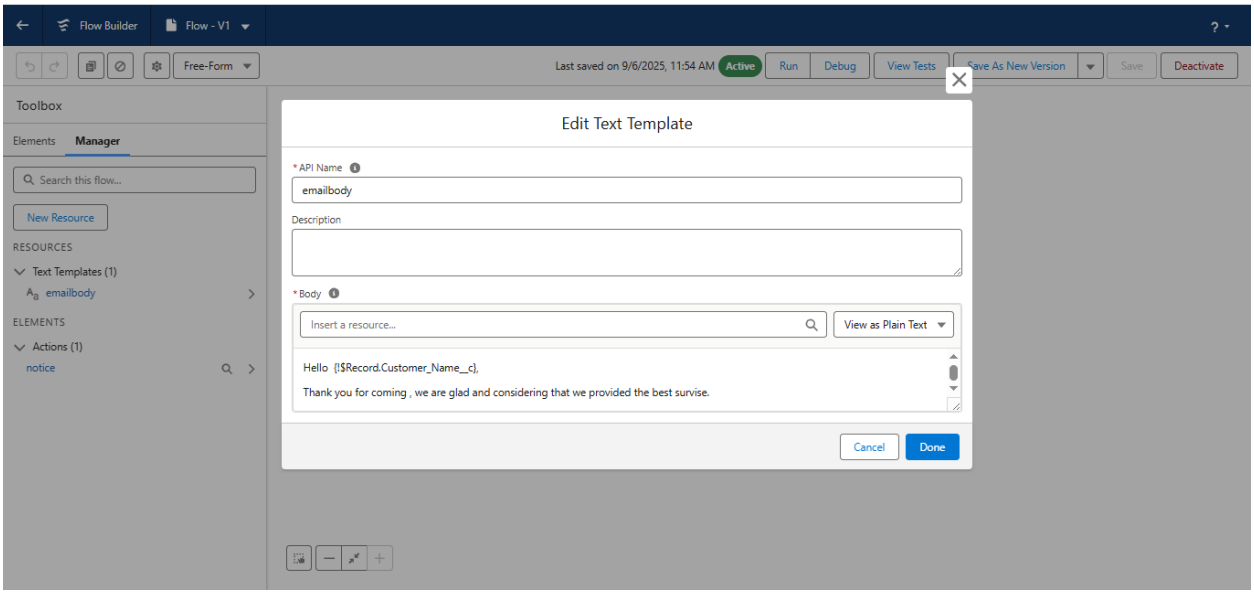

**15.Creating Flow:-**

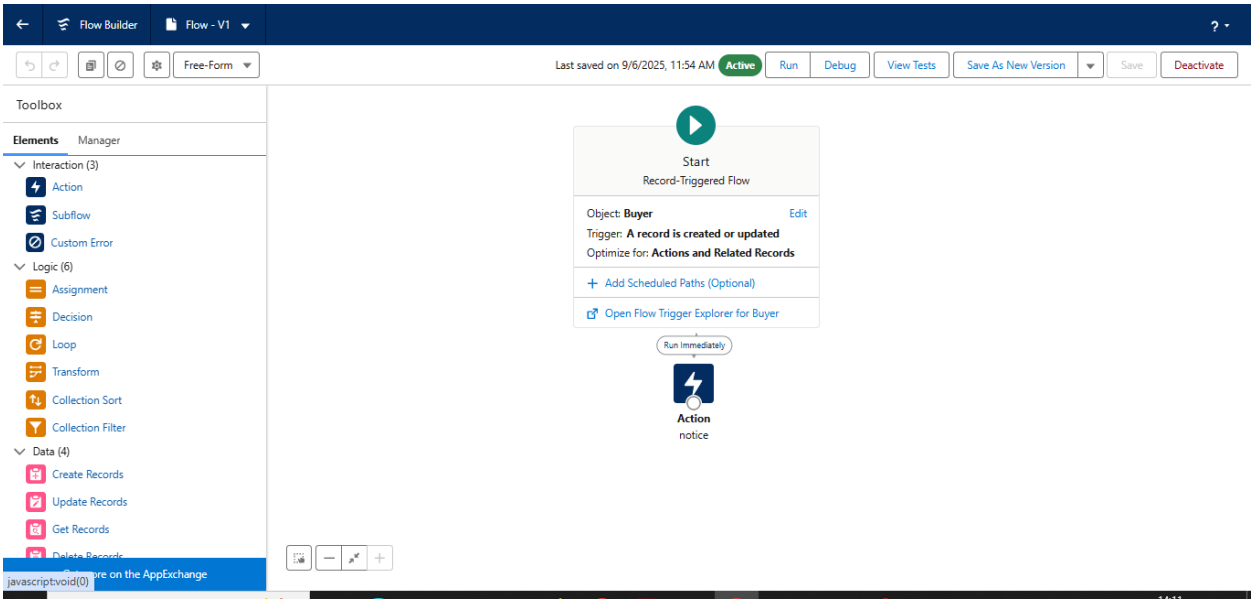
**Purpose:-**

Flows in Salesforce automate processes, saving time and reducing manual work. They also improve communication by ensuring that important actions happen automatically based on system events.

**slaying in Project:-**

- The manager built a record-triggered flow on the Buyer object.
- The flow triggers whenever the manager creates a new record or updates an existing record.
- It auto sends an email receipt to the customer.
- The email includes details such as Customer Name, Amount Paid, Vehicle Type, and Fuel Intake.


Flow Builder
Flow - V1
Free-Form
Last saved on 9/6/2025, 11:54 AM
Active
Run
Debug
View Tests
Save As New Version
Save
Deactivate
Toolbox
Elements
Manager
Search this flow...
New Resource
RESOURCES
Text Templates (1)
emailbody
ELEMENTS
Actions (1)
notice
Edit Text Template
* API Name
emailbody
Description
* Body
Insert a resource...
View as Plain Text
Hello {!$Record.Customer_Name__c},
Thank you for coming , we are glad and considering that we provided the best survise.
Cancel
Done


Flow Builder
Flow - V1
Free-Form
Last saved on 9/6/2025, 11:54 AM
Active
Run
Debug
View Tests
Save As New Version
Save
Deactivate
Toolbox
Elements
Manager
Interaction (3)
Action
Subflow
Custom Error
Logic (6)
Assignment
Decision
Loop
Transform
Collection Sort
Collection Filter
Data (4)
Create Records
Update Records
Get Records
Start
Record-Triggered Flow
Object: Buyer
Edit
Trigger: A record is created or updated
Optimize for: Actions and Related Records
Add Scheduled Paths (Optional)
Open Flow Trigger Explorer for Buyer
Run Immediately
Action
notice
javascript:void(0)

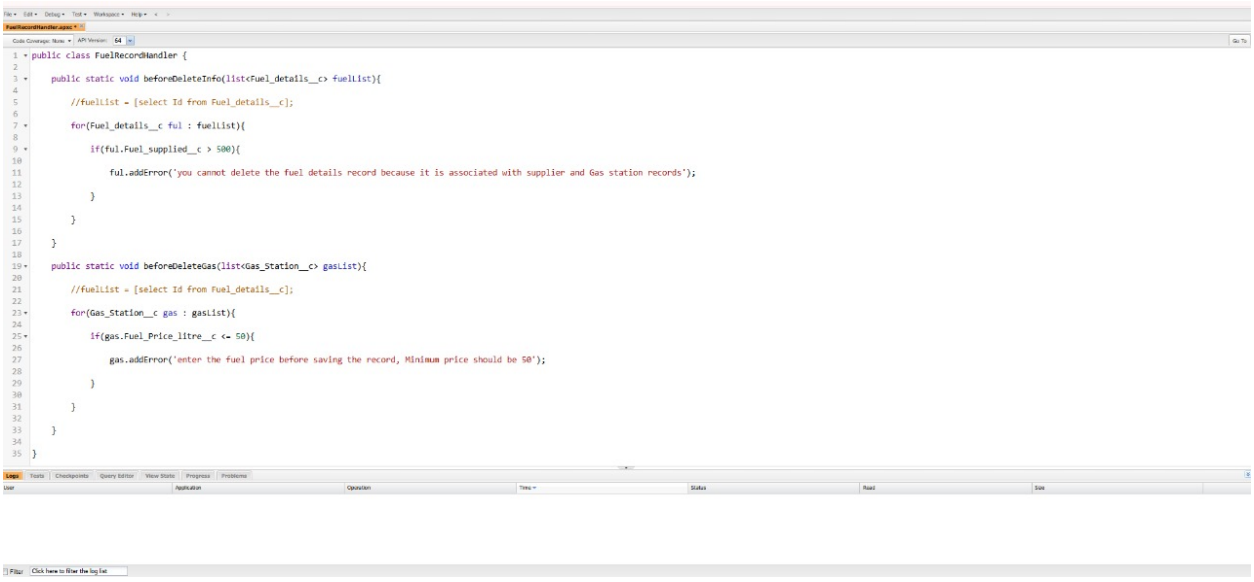
## 16. Creating Apex Triggers:-

**Purpose:-**

Apex triggers automatically execute custom logic during insert, update, or delete events on records. They help enforce business rules and ensure data accuracy.

**performance in Project:-**

- **beforeDeleteInfo** → Prevents deletion of Fuel Details if the supplier quantity is greater than 500.
- **beforeDeleteGas** → Checks that Gas Station fuel price is over 50 before allowing record insertion.



```
public class FuelRecordHandler {

    public static void beforeDeleteInfo(list<Fuel_details__c> fuelList){

        //fuelList = [select Id from Fuel_details__c];

        for(Fuel_details__c ful : fuelList){

            if(ful.Fuel_supplied__c > 500){

                ful.addError('you cannot delete the fuel details record because it is associated with supplier and Gas station records');

            }

        }

    }

    public static void beforeDeleteGas(list<Gas_Station__c> gasList){

        //fuelList = [select Id from Fuel_details__c];

        for(Gas_Station__c gas : gasList){

            if(gas.Fuel_Price_litre__c <= 50){

                gas.addError('enter the fuel price before saving the record, Minimum price should be 50');

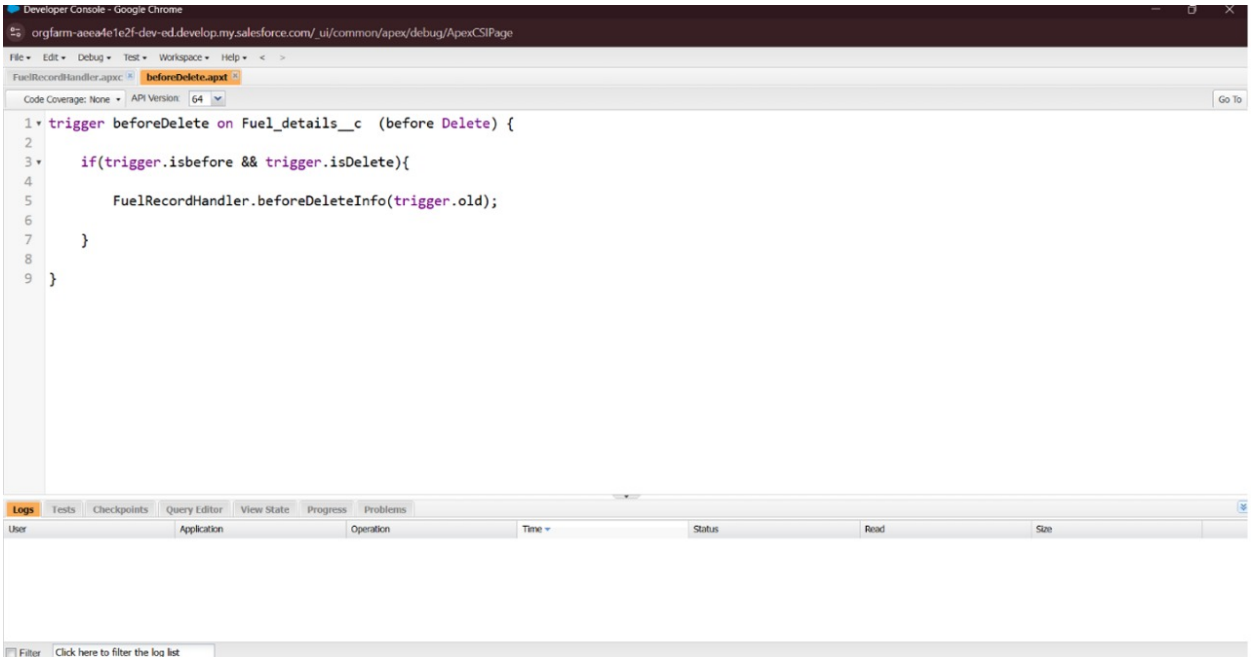
            }

        }

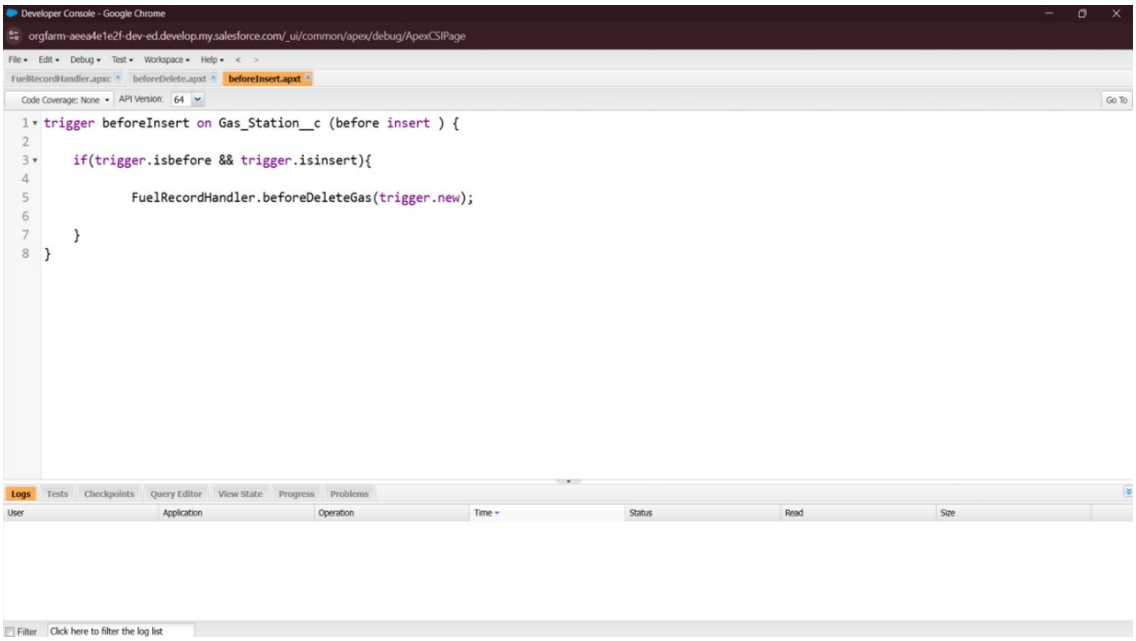
    }

}
```

**Trigger fuel Details (Before Delete)**:-




## Trigger: Gas Station (Before Insert):-



## Feature Enhancements for Gas Station App:-

Salesforce Einstein automates tasks and provides insights. It helps the growth team enhance the Gas Station CRM.

- **Einstein Bots** – Instantly answer customer questions and show fuel prices. They issue receipts automatically and reduce waiting time.
- **Einstein Next Best Action** – Suggests personalized offers or discounts. It uses customer purchase history to boost loyalty and sales.
- **Einstein Analytics Dashboard** – provides visual insights on sales, fuel demand, suppliers, and inventory trends.
- **Einstein Prediction Builder** – Forecasts customer return behavior and fuel demand. It supports loyalty programs and marketing strategies
- **Einstein Activity Capture** – Automatically logs customer interactions to enhance follow-ups and tracking.
- **Einstein Recommendations** – Suggests pricing, promotions, or stock adjustments using historical trends and analytics

These enhancements make the CRM smarter, more customer-focused, and efficient for staff and managers.

## Conclusion:-

The Gas Station CRM in Salesforce streamlines operations by integrating all key processes. It manages buyers, suppliers, fuel transactions, and station details efficiently. Flows and Apex triggers automate tasks and send instant, accurate e-receipts to customers

Custom objects, page layouts, profiles, roles, and permission sets help organize and secure data. Managers, sales executives, and staff have role-based access to view or edit only relevant data. Reports and dashboards provide clear insights into sales, fuel usage, and customer actions. This supports better decision-making and performance tracking.

The CRM system enables AI-driven passport, predictive calculus, and personalized customer offers through Salesforce Einstein. By automating tasks and consolidating operations, it reduces errors and saves time while improving efficiency

The Gas Station CRM makes management smarter and empowers employees. It enhances decisions and offers user-friendly knowledge for staff and customers.